Machine-Learning Screening of Electrocatalytic 5-Hydroxymethylfurfural Oxidation to 2,5-Furandicarboxylic Acid from Literature Data: Leakage-Aware Classification, Regression and Multi-Objective Optimisation

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Abstract

We assembled 154 electrocatalytic HMF-oxidation records from 99 publications and built a leakage-aware machine-learning pipeline that (i) engineers chemical descriptors for the catalyst, electrolyte, cell configuration and operating conditions, (ii) clusters reported conversion and FDCA yield with K-means, (iii) compares six classifier families under four class-balancing strategies, (iv) interprets the selected models with SHAP, (v) regresses four performance outcomes and (vi) searches catalyst/condition space with a validation-gated NSGA-II. All model selection uses publication-grouped cross-validation so that no paper contributes records to both training and test folds. Correcting descriptor extraction (anode-only catalyst parsing, formula-aware compound families, temperature and pH parsing, counter-electrode descriptors), retaining missing-value indicators and re-tuning the models raised the best pooled out-of-fold macro-F1 for a binary $\geq$90% performance screen from 0.586 to 0.731 (neural network with SMOTE; 95% CI 0.717–0.744; stratified dummy 0.480), and for three-cluster K-means labels from 0.421 to 0.626. We also identify and correct an evaluation flaw in which folds where an oversampler could not be fitted were silently discarded, which had inflated earlier $K = 4$ scores. No model–sampler combinations reached a mean macro-F1 of 0.76, reflecting the small, heterogeneous and imbalanced literature corpus rather than a pipeline limitation. Regression models only modestly beat a median baseline, and NSGA-II outputs are presented as hypotheses for experimental testing.

Keywords: HMF electrooxidation; FDCA; machine learning; grouped cross-validation; class imbalance; SHAP; NSGA-II

1 Introduction

2,5-Furandicarboxylic acid (FDCA) is a bio-based replacement for terephthalic acid, and its electrochemical production from 5-hydroxymethylfurfural (HMF) couples biomass valorisation with renewable electricity. Reported performance depends jointly on the anode material, electrolyte, applied potential, temperature, mediator use and cell design, and the published literature now contains enough individual experiments to attempt data-driven screening. Literature-mined datasets are, however, small, heterogeneously reported and dominated by high-performing results, which makes naive machine-learning evaluation prone to optimistic bias: records from the same paper share catalysts and protocols, so random cross-validation leaks information between training and test sets.

Following the integrated workflow of Li *et al.* [1], which combined K-means labelling, oversampling, classifier comparison, SHAP interpretation and a genetic algorithm for thermal aerobic HMF oxidation, we adapt the methodology to electro-oxidation and make three contributions:

(1) a publication-grouped, fold-local evaluation protocol in which imputation, scaling and oversampling are fitted only on training papers; (2) a corrected descriptor set for the anode catalyst, electrolyte and cell configuration; and (3) an honest assessment of how far classification, regression and optimisation can be pushed with the current corpus.

2 Data and methods

2.1 Literature dataset

The corpus contains 154 experiments from 99 publications (Table 1). Each record lists the cell type, HMF concentration, catalyst description, electrolyte, applied potential, TEMPO use, temperature and up to four outcomes: HMF conversion, FDCA yield, faradaic efficiency (FE) and selectivity. Outcome reporting is incomplete; only 60 records report all four outcomes.

Table 1: Dataset composition. Missing counts per continuous descriptor: `Temperature_C` (63), `Potential_vs_RHE_V` (32), `Electrolyte_Concentration_M` (10), `HMF_Concentration_mM` (6), `Electrolyte_pH` (2).

Quantity	Value
Literature records	154
Independent publications (CV groups)	99
Input descriptors (before missing-value indicators)	61
Missing-value indicator columns added in-fold	5
Records reporting conversion	117
Records reporting FDCA yield	123
Records reporting faradaic efficiency	119
Records reporting selectivity	91
Records reporting all four outcomes	60
Binary-screen records (FDCA product, yield reported)	119 (89 positive)

2.2 Descriptor engineering

Inputs are restricted to eight agreed experimental groups: TEMPO, temperature, reactant, HMF concentration, catalyst, electrolyte, potential versus RHE and cell configuration. All measured outcomes and any column derived from them are excluded. Three corrections to the earlier descriptor extraction were made in this revision.

- **Anode-only catalyst parsing.** Catalyst strings describe the whole cell (e.g. "Anode: Ni_3N@C on Ni foam. Cathode: Pt foil"). The previous parser searched the full string, so 74 records were flagged as Pt catalysts because of the Pt counter electrode. Element and compound flags are now extracted from the anode/working-electrode segment only, which leaves 2 genuine Pt anodes.
- **Formula-aware compound families.** Phosphide, nitride, sulfide and selenide flags had been matched only on words such as "phosphide" and were zero for every record. Case-sensitive formula patterns (Ni_3S_2, CoNiP, VN, Co_9S_8, Ni_2P) now recover 12 phosphide, 8 nitride and 15 sulfide anodes, plus borides, foam substrates, conductive supports (FTO, ITO, BDD), element counts, a noble-metal flag and a bimetallic flag.
- **Operating conditions.** "$23 \pm 2\,^{\circ}C$" had been parsed as $2\,^{\circ}C$; uncertainties are now stripped first. An electrolyte pH is taken from explicit "pH" values or inferred as $14 + \log_{10}[OH^-]$

for hydroxides. Counter-electrode descriptors (Pt, carbon, Ni, reported) are added to the cell-configuration group.

This yields 61 descriptors. Within every training fold, missing continuous values are median-imputed and a binary indicator is appended for each of the 5 descriptors with missing values. Whether a paper reports, for example, its temperature or potential is a property of the input record and carries information that median imputation alone would erase.

2.3 Classification targets

K-means labels. Conversion and FDCA yield are standardised and clustered for $K \in \{3, 4, 5, 6\}$ (100 initialisations). Missing outcomes are median-imputed for clustering only and flagged; labels are never used as inputs (Table 2).

Binary performance screen. For records whose major product is FDCA and whose yield is reported, a performance score

$$P = \begin{cases} \min(X, Y) & \text{if conversion } X \text{ is reported,} \\ Y & \text{otherwise,} \end{cases} \tag{1}$$

is computed from conversion X and yield Y, and records with $P \geq 90\%$ form the positive class.

Table 2: K-means cluster sizes on (conversion, FDCA yield). The minority clusters are extremely small, which bounds the attainable macro-F1 under publication-grouped validation.

K	Cluster sizes	Smallest cluster
3	138, 5, 11	5
4	137, 3, 4, 10	3
5	17, 7, 2, 4, 124	2
6	127, 2, 4, 6, 14, 1	1

2.4 Models and class balancing

Six classifier families were compared: decision tree (DT), RBF support vector machine (SVM), random forest (RF), multilayer perceptron (NN), gradient boosting (GB) and extremely randomised trees (ET), each under no balancing, random oversampling (ROS), SMOTE [2] and ADASYN [3], using scikit-learn [4] and imbalanced-learn [5]. Hyperparameters were selected from a small per-family grid evaluated with the same grouped protocol: DT depth 6; SVM $C = 0.3$, $\gamma = 0.01$; RF 300 trees of depth 6; NN one hidden layer of 32 units with L_2 penalty $\alpha = 0.1$ and *no* early stopping (the former early-stopping split removed 10% of an already tiny training fold and often converged on the majority class); GB 100 depth-1 stumps; ET 400 trees with minimum leaf size 2. Class-weighted losses are used where supported.

2.5 Evaluation protocol

All scores come from 25 repeats of 5-fold `StratifiedGroupKFold` with publication as the group. Imputation, indicator creation, variance filtering, scaling and oversampling are fitted on the training papers of each fold only. Within a repeat, out-of-fold predictions from all folds are pooled and a single macro-F1 is computed,

$$\mathrm{F1}_{\mathrm{macro}} = \frac{1}{C} \sum_{c=1}^{C} \frac{2\,\mathrm{P}_c\,\mathrm{R}_c}{\mathrm{P}_c + \mathrm{R}_c}, \tag{2}$$

where P_c and R_c are the precision and recall of class c. The mean, standard deviation and t-based 95% confidence interval are reported over repeats. The selected model is compared with a stratified dummy classifier on identical splits and with a permutation test in which labels are shuffled within publications.

Correction of fold-dropping bias

SMOTE and ADASYN require more minority-class training rows than neighbours. In the previous implementation a fold that failed this requirement was silently skipped and the repeat's macro-F1 was computed on the remaining folds; for $K = 4$ only about 1.2 of 5 folds (35 of 154 records) were scored. Neighbour counts now adapt to $\min(2, n_{\min} - 1)$, a repeat counts only if every fold produced predictions, and a combination is ranked only if at least 80% of its repeats are complete. 30 combinations (mostly SMOTE/ADASYN at $K \geq 5$, where single-record classes occur) are therefore reported as infeasible rather than scored.

2.6 Interpretation, regression and optimisation

SHAP values [6] are computed on a held-out 20% of publications, with Tree SHAP for tree ensembles and Kernel SHAP through the complete fitted pipeline otherwise. For the K-means analysis the best Stage-04 model for each K is explained. Conversion, yield, FE and selectivity are regressed per target (random forest with mutual-information feature selection, $k = 15$) on all records that report that target, using 10 repeats of 3-fold GroupKFold; a joint four-output extra-trees model on complete rows is reported as a secondary analysis. A validation gate requires every target to beat the median baseline in mean MAE and in a majority of folds before the regressors may drive optimisation.

NSGA-II [7] searches a mixed binary/continuous encoding of the descriptor space, maximising predicted conversion, yield and FE while minimising temperature and potential. Pareto solutions are ranked by an entropy-weighted fitness

$$F = w_1\tilde{X} + w_2\tilde{Y} + w_3\widetilde{\mathrm{FE}} - w_4\tilde{T} - w_5\tilde{E}, \qquad w_j = \frac{1 - e_j}{\sum_k (1 - e_k)}, \tag{3}$$

with min–max scaled objectives and e_j the Shannon entropy of objective j over the front. Convergence is tracked with the spacing metric of the first front.

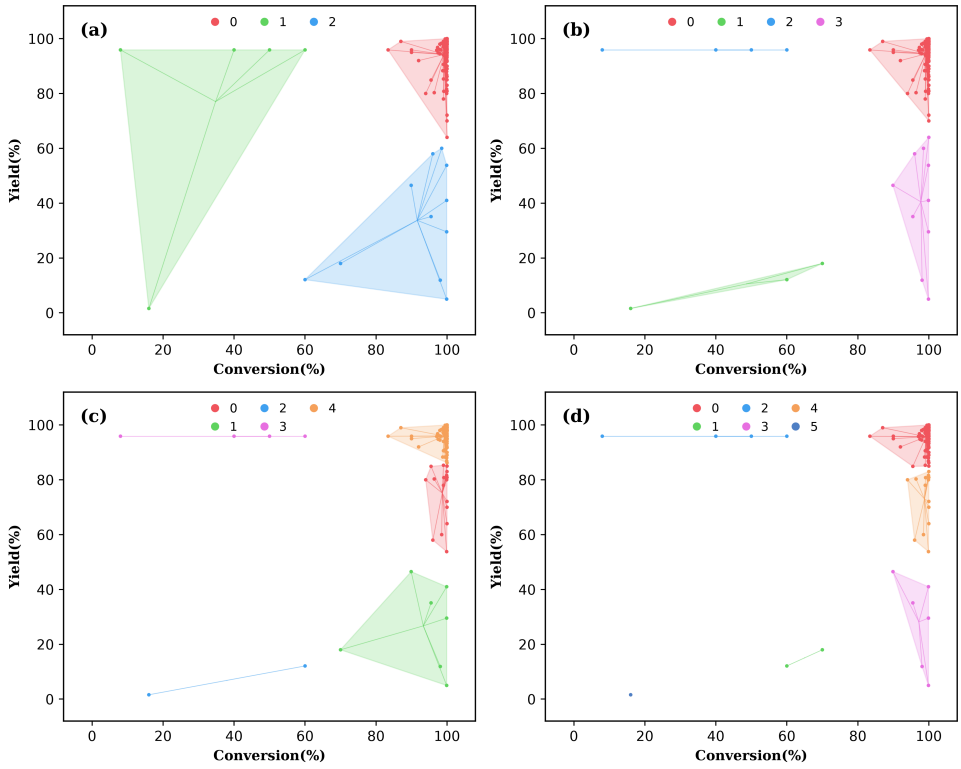
3 Results

3.1 Clustering

K-means separates a dominant high-conversion/high-yield group from small groups with low yield at high conversion or low conversion (Figure 1). 52 of 154 records required outcome imputation for clustering, and the smallest clusters contain between one and five records (Table 2). Macro-F1 weights each class equally, so misclassifying a two-record cluster costs as much as misclassifying the majority class.

3.2 Classification performance

Table 3 summarises the effect of the revision. For the binary screen (119 records, 89 positive), the best model is the neural network with SMOTE (macro-F1 0.731 ± 0.033), against 0.480 for a stratified dummy classifier (Table 4, Figure 3). Extra-trees without balancing is statistically indistinguishable (0.720). Every pre-existing family improved on the binary screen, by +0.04 (DT) to +0.22 (NN). For $K = 3$ the best model is GB + none (0.626; Table 5). For $K \geq 4$, scores



Without sampling K-means clustering results: (a) K = 3, (b) K = 4, (c) K = 5, and (d) K = 6

Figure 1: K-means clustering of reported conversion and FDCA yield for $K = 3$–6.

stay between 0.2 and 0.45: several classes have fewer than five members, and a single grouped fold frequently contains none of them in training. The within-publication permutation test gives $p = 0.47$. Because most descriptors are constant within a paper, shuffling labels inside papers preserves the paper-level signal the model exploits, so this test is conservative here; the dummy comparison and the confidence interval are the more informative checks.

3.3 Learning curves

Training macro-F1 stays well above validation macro-F1 for every model and K (Table 6, Figure 7), indicating variance-limited models. At $K = 3$ the SVM validation curve is still rising at the largest training size and the other families fluctuate without a clear plateau, so additional publications are the most direct route to higher scores; for $K \geq 4$ the curves are flat because the minority clusters do not grow with more records.

3.4 Model interpretation

Table 7 and Figures 9–11 list the descriptors with the largest SHAP contributions. Support type (conductive glass versus foam), copper and iron content, oxyhydroxide/hydroxide phases, counter-electrode material, potential and the temperature-not-reported indicator recur across models. These rankings describe what the models use to separate the literature records; they partly encode publication style and are not evidence of causal catalyst effects.

3.5 Regression

Per-target models reduce MAE relative to the median baseline for all four outcomes (Table 8), but R^2 remains between -1.73 and 0.04: the outcome distributions are concentrated near 100% with

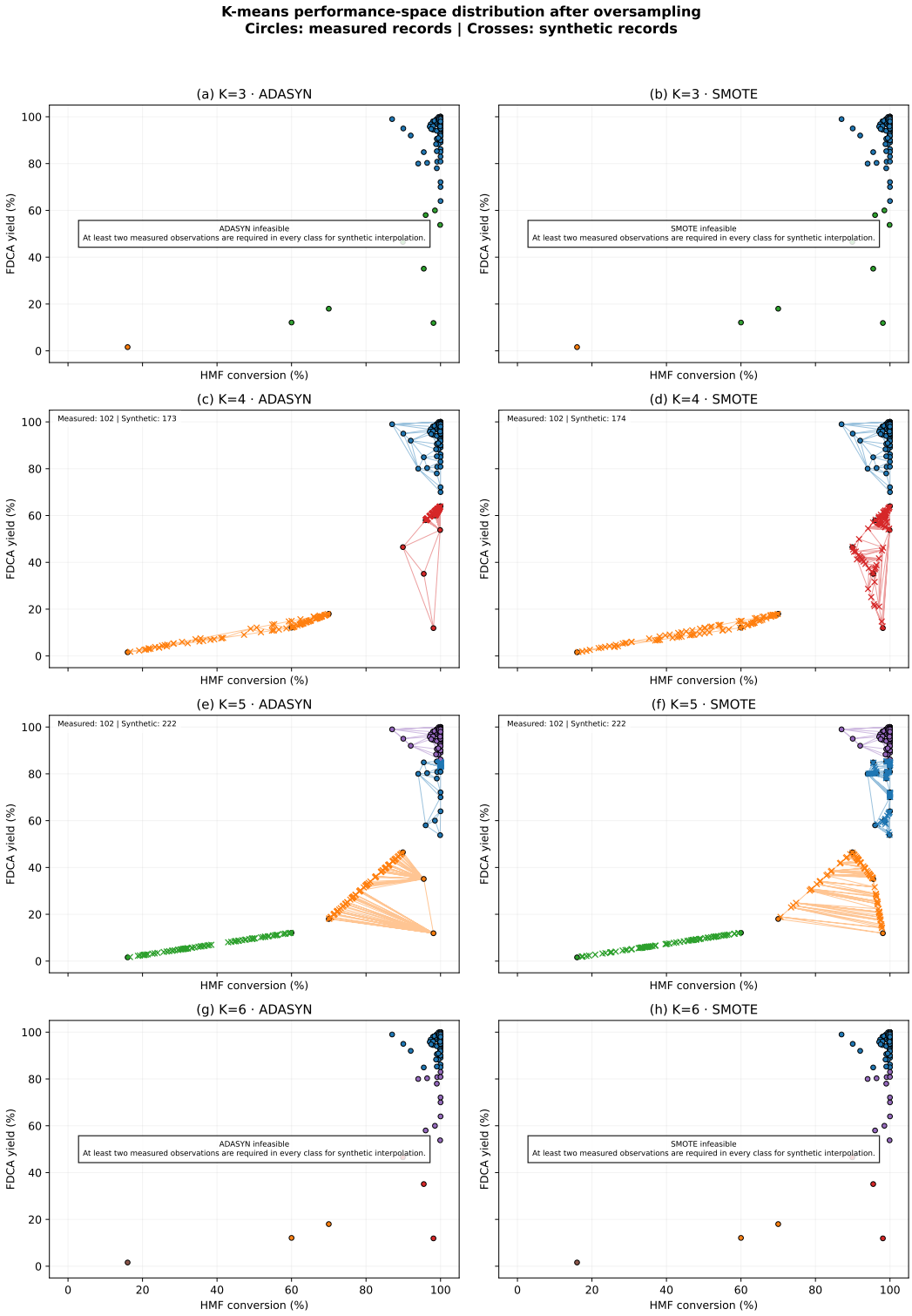


Figure 2: ADASYN and SMOTE synthetic records (crosses) in performance space for $K = 3$–6, drawn for the 102 records with both outcomes measured. Panels marked infeasible contain a class with fewer than two measured records.

Table 3: Best pooled out-of-fold macro-F1 per classifier family and task (best sampler in parentheses), before (commit `bdbd2e2`) and after the descriptor, preprocessing, model and evaluation revisions. 95% CI over 25 repeats. Earlier $K = 4$ values were computed on the subset of folds where SMOTE/ADASYN could be fitted and are therefore not comparable (Section 2.5).

Task	Model	Before	After	95% CI	Δ
Binary ≥90%	DT	0.557 (ROS)	0.600 (ADASYN)	[0.580, 0.619]	+0.042
	SVM	0.534 (ADASYN)	0.695 (ROS)	[0.671, 0.718]	+0.160
	RF	0.586 (ROS)	0.678 (ROS)	[0.659, 0.698]	+0.092
	NN	0.507 (ADASYN)	0.731 (SMOTE)	[0.717, 0.744]	+0.224
	GB	0.575 (ROS)	0.661 (SMOTE)	[0.641, 0.681]	+0.086
	ET	–	0.720 (none)	[0.706, 0.733]	new
K-means, $K = 3$	DT	0.410 (ROS)	0.519 (none)	[0.487, 0.550]	+0.109
	SVM	0.298 (ADASYN)	0.486 (none)	[0.448, 0.524]	+0.187
	RF	0.379 (ROS)	0.481 (ROS)	[0.443, 0.518]	+0.102
	NN	0.389 (SMOTE)	0.535 (ROS)	[0.496, 0.574]	+0.146
	GB	0.421 (ROS)	0.626 (none)	[0.601, 0.650]	+0.205
	ET	–	0.484 (SMOTE)	[0.474, 0.495]	new
K-means, $K = 4$	DT	0.360 (SMOTE)	0.427 (SMOTE)	[0.393, 0.462]	+0.068
	SVM	0.458 (SMOTE)	0.290 (none)	[0.277, 0.302]	-0.168
	RF	0.477 (ADASYN)	0.290 (ROS)	[0.281, 0.300]	-0.187
	NN	0.248 (ADASYN)	0.318 (SMOTE)	[0.300, 0.337]	+0.071
	GB	0.437 (SMOTE)	0.314 (SMOTE)	[0.306, 0.323]	-0.123
	ET	–	0.325 (SMOTE)	[0.318, 0.331]	new
K-means, $K = 5$	DT	0.232 (ROS)	0.220 (ROS)	[0.211, 0.229]	-0.012
	SVM	0.255 (none)	0.302 (ROS)	[0.296, 0.307]	+0.047
	RF	0.272 (ROS)	0.286 (ROS)	[0.279, 0.293]	+0.014
	NN	0.237 (ROS)	0.321 (ROS)	[0.307, 0.335]	+0.084
	GB	0.265 (ROS)	0.278 (none)	[0.268, 0.289]	+0.014
	ET	–	0.298 (none)	[0.286, 0.310]	new
K-means, $K = 6$	DT	0.215 (ROS)	0.225 (none)	[0.212, 0.238]	+0.010
	SVM	0.215 (none)	0.245 (ROS)	[0.238, 0.253]	+0.030
	RF	0.215 (ROS)	0.250 (ROS)	[0.246, 0.255]	+0.035
	NN	0.198 (ROS)	0.283 (ROS)	[0.268, 0.299]	+0.085
	GB	0.207 (none)	0.231 (ROS)	[0.225, 0.237]	+0.024
	ET	–	0.254 (ROS)	[0.247, 0.261]	new

Table 4: Binary ≥90% screen: pooled out-of-fold macro-F1 (mean ± SD over 25 repeats) for every model and sampler.

Model	None	ROS	SMOTE	ADASYN
DT	0.593 ± 0.031	0.593 ± 0.040	0.598 ± 0.042	0.600 ± 0.047
SVM	0.681 ± 0.032	0.695 ± 0.056	0.630 ± 0.043	0.637 ± 0.026
RF	0.658 ± 0.042	0.678 ± 0.047	0.640 ± 0.036	0.630 ± 0.027
NN	0.718 ± 0.038	0.718 ± 0.037	**0.731 ± 0.033**	0.721 ± 0.035
GB	0.568 ± 0.033	0.631 ± 0.045	0.661 ± 0.048	0.645 ± 0.047
ET	0.720 ± 0.032	0.696 ± 0.041	0.716 ± 0.038	0.713 ± 0.030

Table 5: K-means $K = 3$: pooled out-of-fold macro-F1 (mean ± SD over 25 repeats).

Model	None	ROS	SMOTE	ADASYN
DT	0.519 ± 0.076	0.512 ± 0.079	0.450 ± 0.056	0.454 ± 0.065
SVM	0.486 ± 0.093	0.403 ± 0.102	0.436 ± 0.099	0.422 ± 0.085
RF	0.328 ± 0.024	0.481 ± 0.091	0.358 ± 0.037	0.360 ± 0.040
NN	0.438 ± 0.073	0.535 ± 0.095	0.515 ± 0.084	0.520 ± 0.084
GB	**0.626 ± 0.060**	0.591 ± 0.043	0.604 ± 0.081	0.603 ± 0.097
ET	0.457 ± 0.047	0.444 ± 0.041	0.484 ± 0.026	0.478 ± 0.027

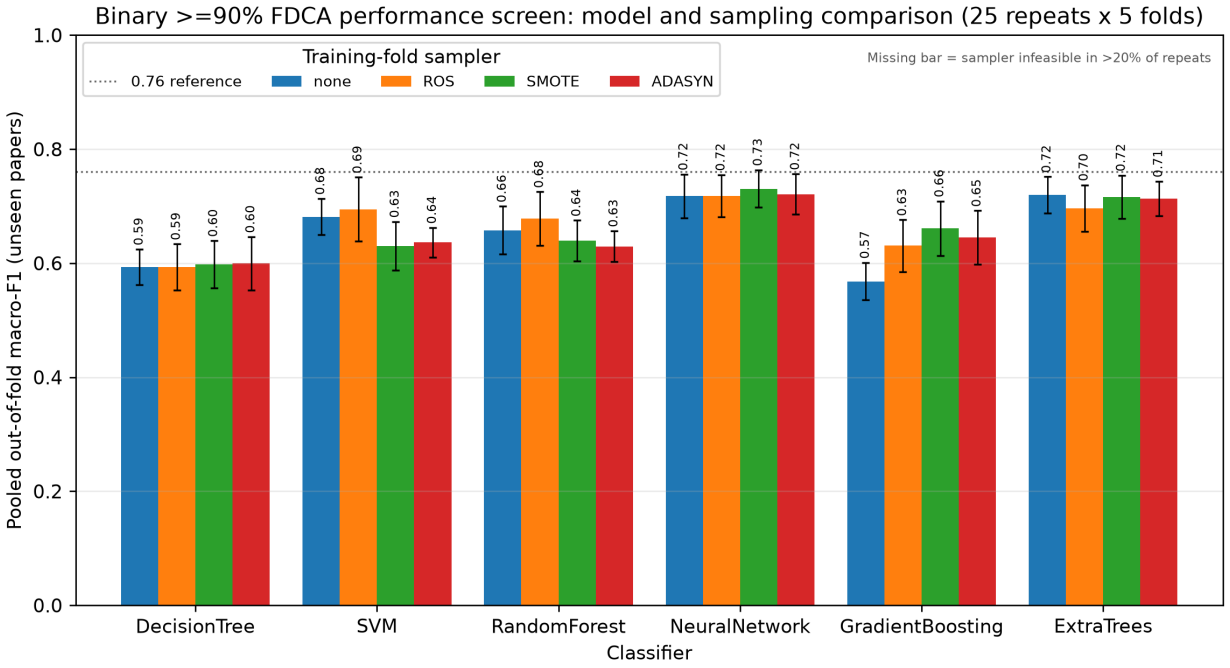


Figure 3: Binary ≥90% screen: pooled out-of-fold macro-F1 by classifier and training-fold sampler (error bars: SD over 25 repeats; dotted line: 0.76).

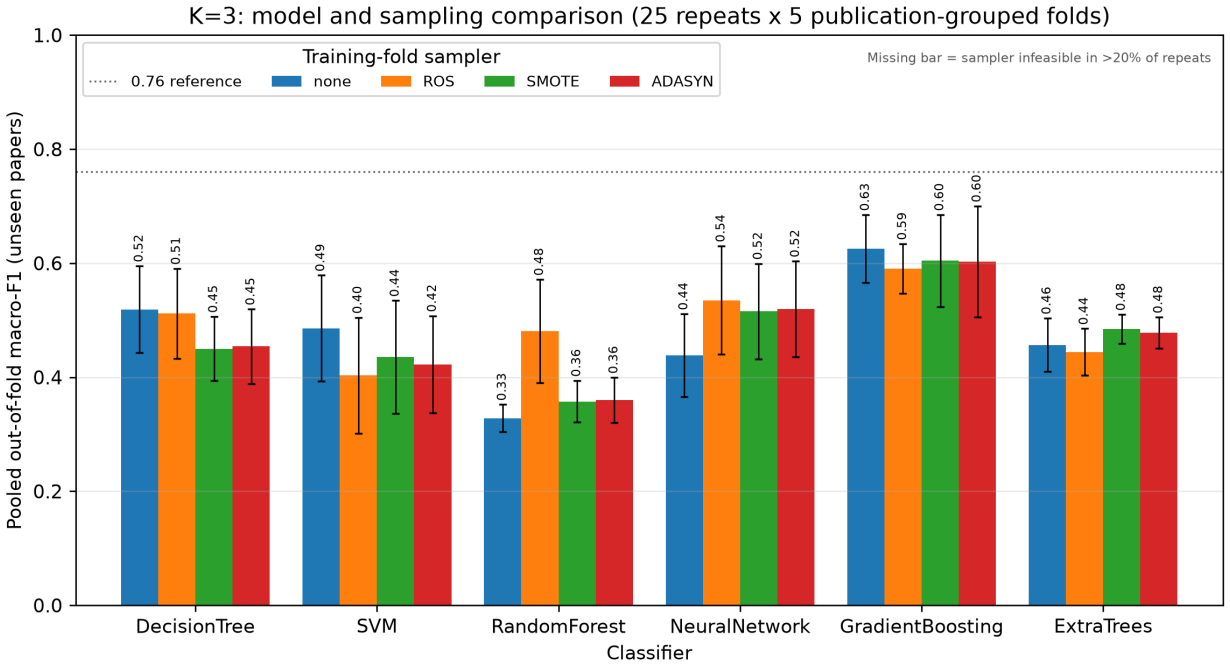


Figure 4: K-means $K = 3$: pooled out-of-fold macro-F1 by classifier and sampler.

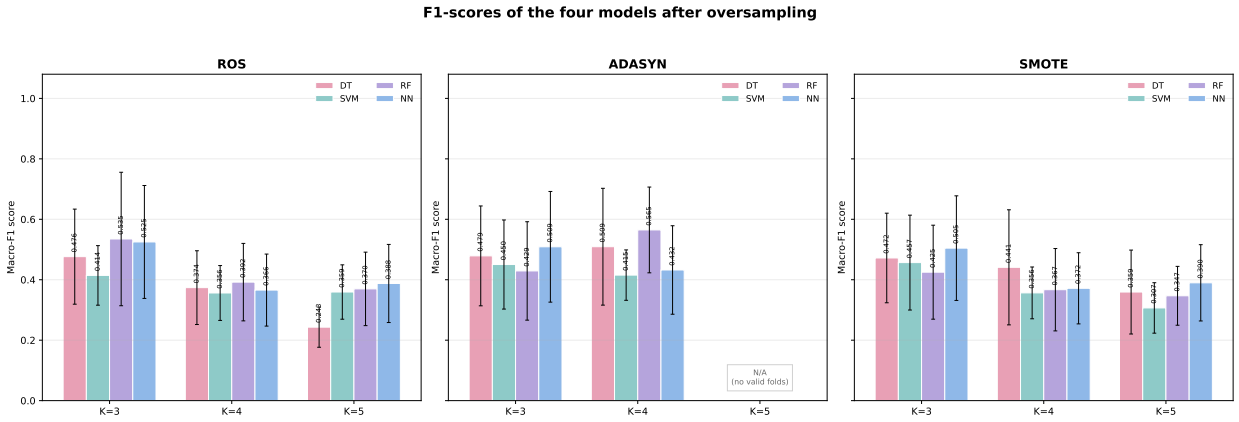


Figure 5: Fold-averaged macro-F1 of DT, SVM, RF and NN after ROS, ADASYN and SMOTE for $K = 3$–5 (reference-paper layout). Fold averages are computed over feasible folds only and differ from the pooled scores in Table 3.

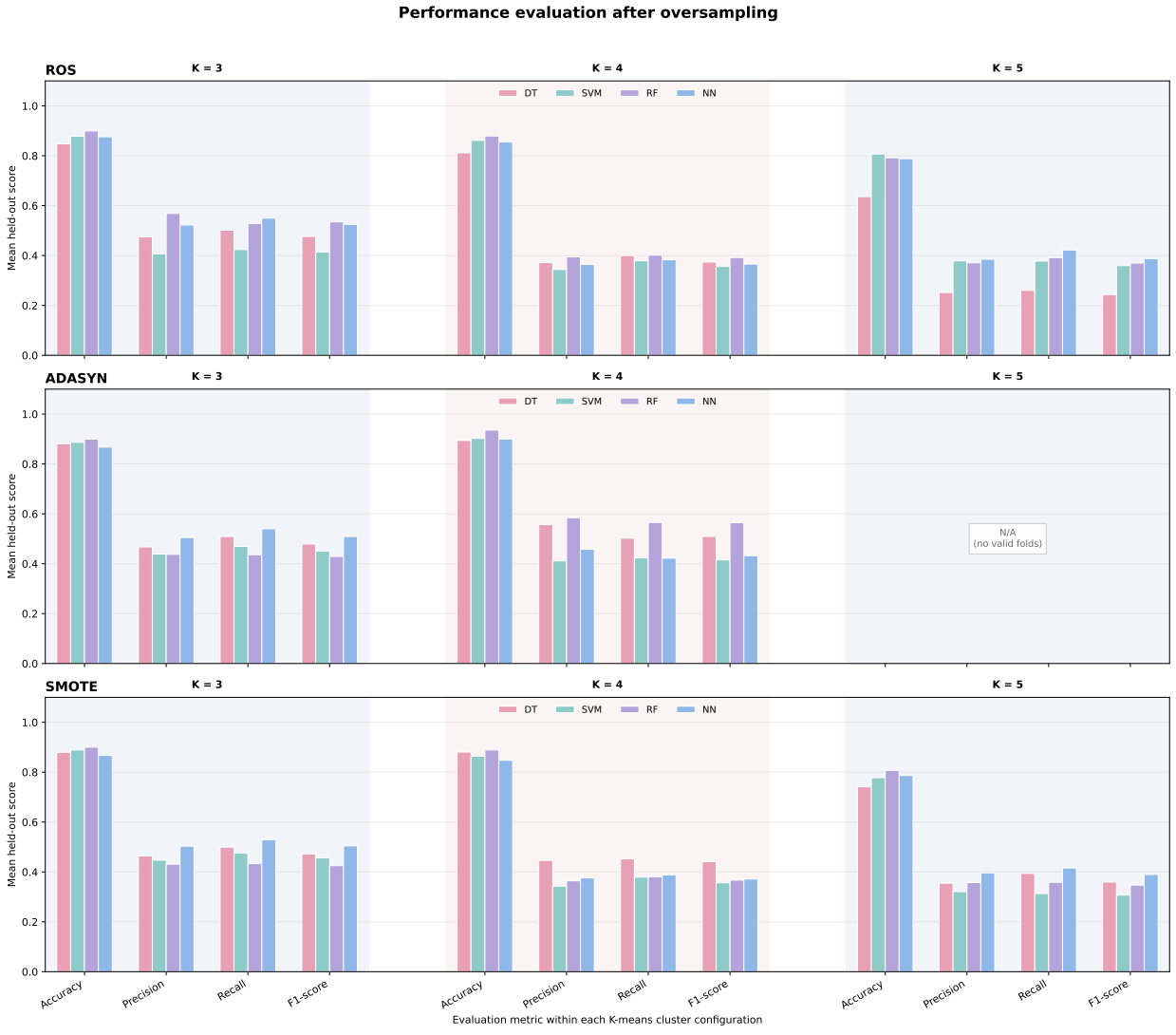


Figure 6: Accuracy, precision, recall and F1 after oversampling.

Table 6: Learning-curve end points (full training fraction, publication-grouped 80/20 holdouts, 10 repeats).

K	Model	Train macro-F1	Validation macro-F1	Gap
3	DT	0.920	0.481	0.438
3	NN	0.959	0.537	0.421
3	RF	0.939	0.470	0.469
3	SVM	0.834	0.729	0.105
4	DT	0.556	0.286	0.269
4	NN	0.969	0.299	0.670
4	RF	0.959	0.228	0.731
4	SVM	0.909	0.223	0.685
5	DT	0.740	0.209	0.532
5	NN	0.774	0.315	0.458
5	RF	0.874	0.231	0.642
5	SVM	0.734	0.250	0.483

Table 7: Top-ranked descriptors by mean absolute SHAP value on held-out publications. Values are associational, not causal; `missingindicator_*` columns flag unreported conditions.

Binary screen (NN + SMOTE)		$K = 3$ (GB + none)		$K = 5$ (NN + ROS)	
`Catalyst_Conductive_Glass`	0.104	`Catalyst_Conductive_Glass`	0.096	`Catalyst_Pd`	0.019
`Catalyst_Oxyhydroxide`	0.053	`Potential_vs_RHE_V`	0.023	`Catalyst_Conductive_Glass`	0.013
`Counter_Electrode_Carbon`	0.045	`HMF_Concentration_mM`	0.010	`Divided_Cell`	0.011
`Catalyst_V`	0.044	`Catalyst_Metal`	0.006	`Catalyst_Fe`	0.010
`Catalyst_Cu`	0.042	`Catalyst_Carbon`	0.006	`Catalyst_N_Elements`	0.009
`Catalyst_Fe`	0.033	`Catalyst_Cu`	0.005	`Potential_vs_RHE_V`	0.009
`Catalyst_Hydroxide`	0.031	`Catalyst_Fe`	0.004	`Catalyst_Hydroxide`	0.008
`missingindicator_Temperature_C`	0.021	`Three_Electrode`	0.004	`Temperature_C`	0.007
`Catalyst_Carbon`	0.020	`Electrolyte_Concentration_M`	0.003	`Catalyst_Sulfide`	0.006
`Catalyst_Bimetallic`	0.019	`Temperature_C`	0.002	`Catalyst_Phosphide`	0.005

a few low outliers, so the models mainly learn to avoid large errors on the typical record rather than to explain variance. The validation gate passed. The joint model trained on the 60 complete records does not beat the baseline (Table 9).

Table 8: Per-target regression (primary): publication-grouped cross-validation against a median baseline. MAE and RMSE in percentage points.

Target	R^2	MAE	Baseline MAE	RMSE	Beats baseline
Conversion	-1.733	6.41	7.07	13.51	yes
Faradaic efficiency	-0.066	8.60	9.95	14.97	yes
Selectivity	0.030	9.97	12.37	17.20	yes
FDCA yield	0.044	11.86	13.19	19.50	yes

Table 9: Joint four-output regression on complete rows (secondary): publication-grouped cross-validation against a median baseline. MAE and RMSE in percentage points.

Target	R^2	MAE	Baseline MAE	RMSE	Beats baseline
Conversion	-3.579	2.55	2.08	7.67	no
Faradaic efficiency	-0.376	7.28	6.89	12.43	no
Selectivity	-0.021	5.82	5.34	10.98	no
FDCA yield	-0.049	5.74	5.29	11.11	no

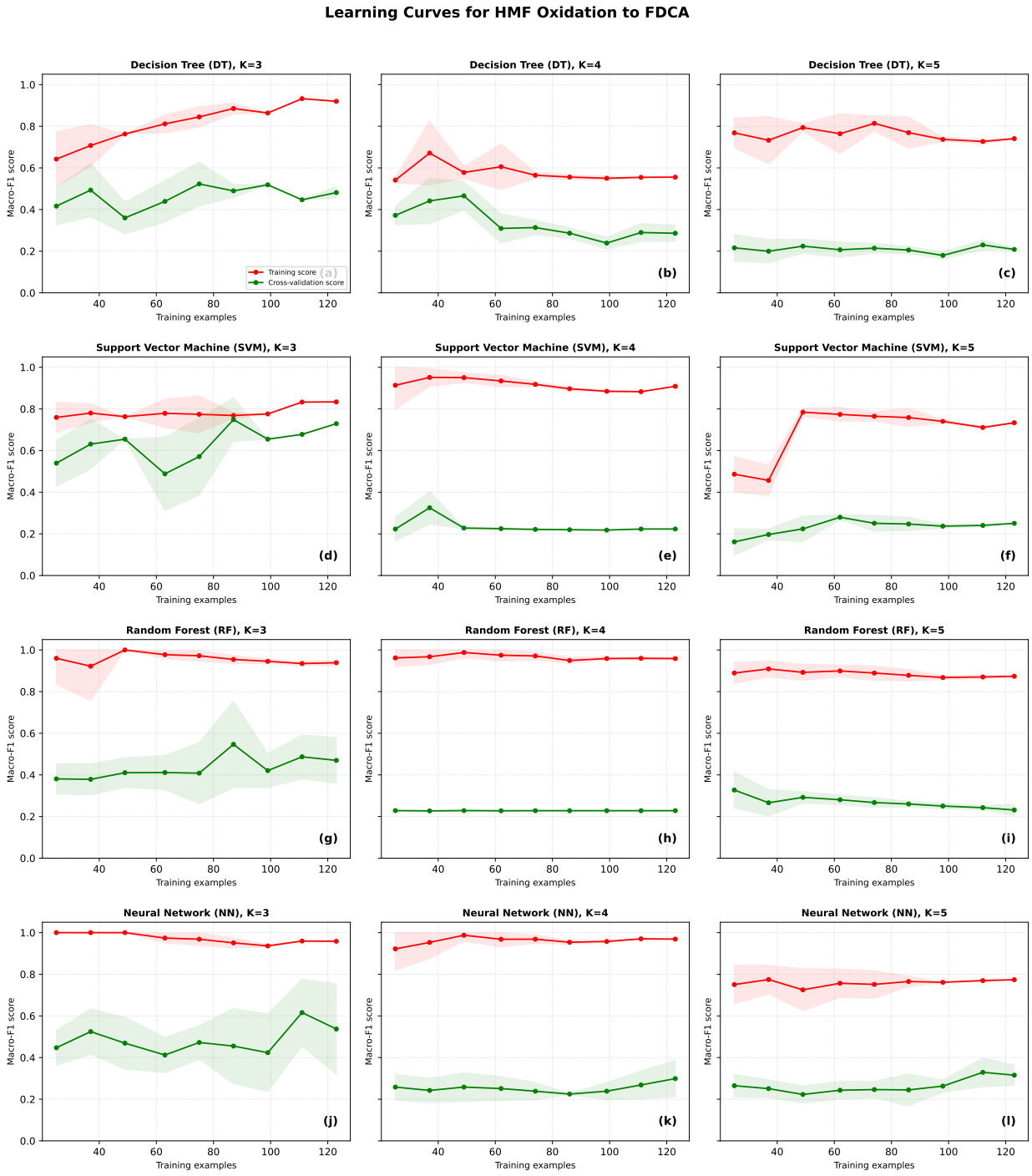


Figure 7: Learning curves (macro-F1) for DT, SVM, RF and NN at $K = 3$, 4 and 5 with publication-grouped holdouts; shading shows ±1 SD.

3.6 Multi-objective optimisation

NSGA-II (100 individuals, 1000 generations) returned 96 Pareto candidates. Entropy weights for the unconstrained search were: Conversion 0.10, FDCA yield 0.14, Faradaic efficiency 0.18, temperature 0.13, potential 0.45. In the constrained $Ni(OH)_2/Cu(OH)_2$ search, all ten top-ranked solutions are $Ni(OH)_2$ on foam, at 0.42–1.10 M KOH, 1.41–1.42 V vs RHE and 25–26 °C (Table 10, Figure 12). The top candidate's nearest literature record (Ni(OH)2/NF, 1.0 M KOH | 1.45 V vs. RHE) reported 88.3% yield against 95.2% predicted; that difference is smaller than the yield regression MAE of 11.9 percentage points. The best $Cu(OH)_2$ candidate is predicted at 91.0%

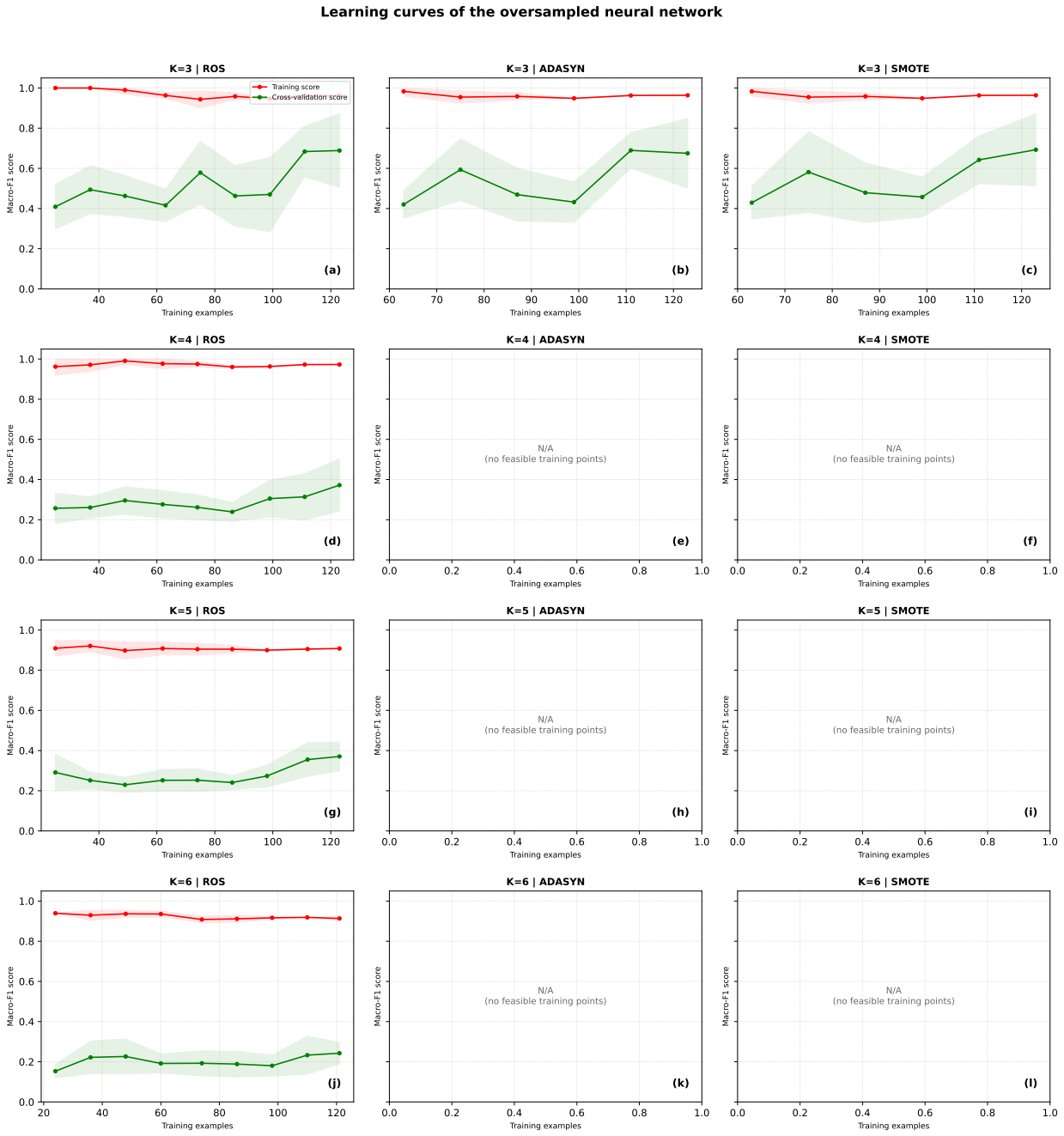


Figure 8: Neural-network learning curves with ROS, ADASYN and SMOTE applied to training folds only.

yield and 94.1% FE. The spacing metric settles into a stable band after about 20 generations and fluctuates around it thereafter (Figure 13). Because the predicted gains lie within the regression error, the candidates should be read as experimental starting points, not predictions of record performance.

4 Discussion

What improved the scores. Most of the gain came from data quality rather than model capacity. Removing counter-electrode contamination and recovering compound families gave the models chemically meaningful catalyst descriptors; the missing-value indicators restored information about reporting practice; and disabling early stopping stopped the neural network from converging on the majority class. Tuning added smaller, family-specific gains.

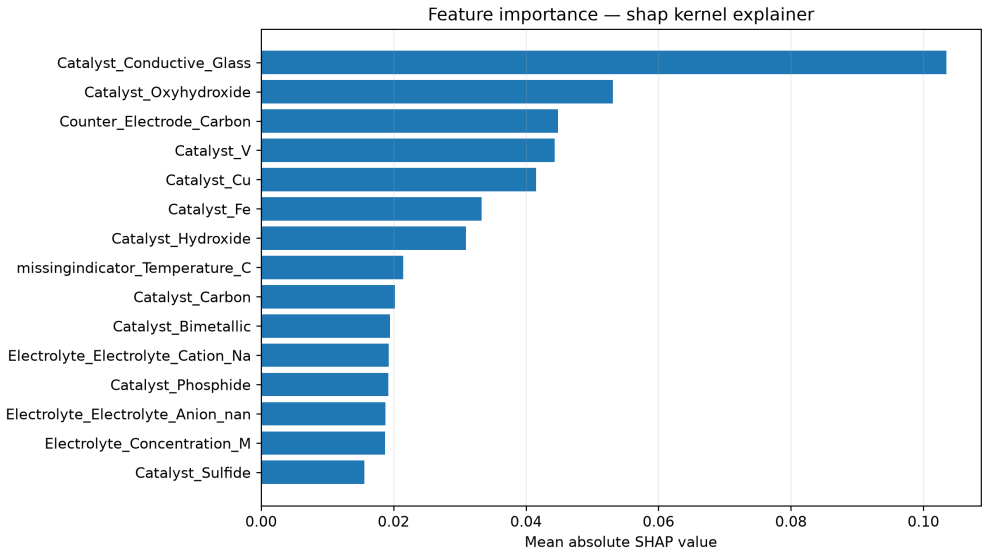


Figure 9: SHAP feature importance for the selected binary-screen model.

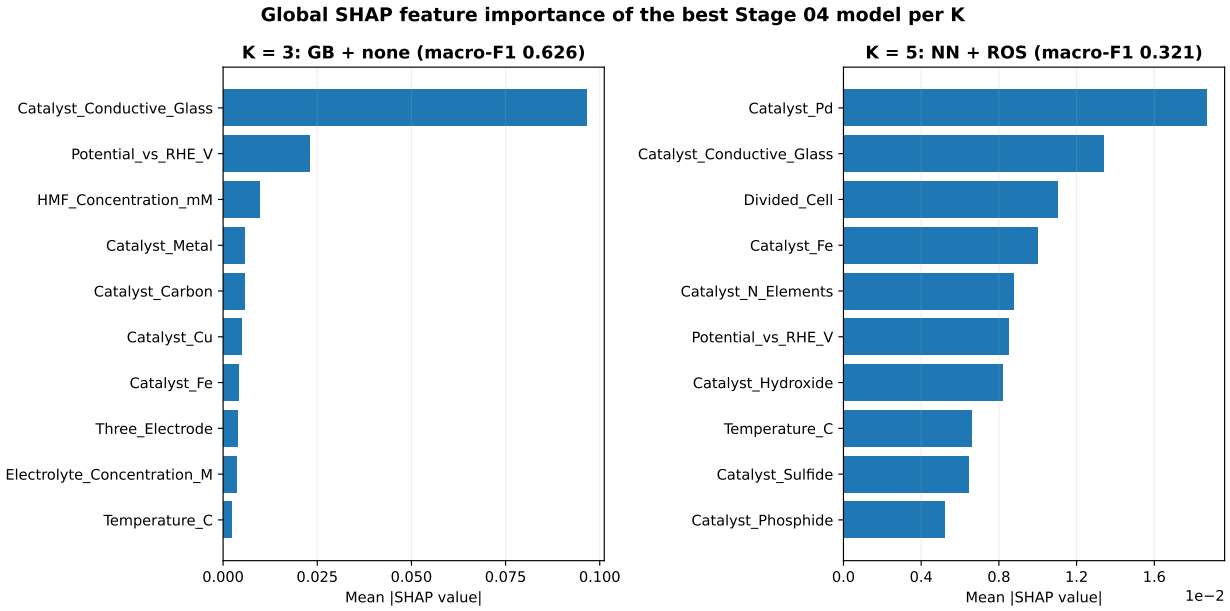


Figure 10: Global SHAP importance of the best Stage-04 model for $K = 3$ and $K = 5$.

Table 10: NSGA-II top-10 candidates in the constrained $Ni(OH)_2/Cu(OH)_2$ search, ranked by entropy-weighted fitness. All performance values are model predictions, not measurements.

Rank	Catalyst	Electrolyte	HMF (mM)	E (V vs RHE)	T (°C)	Conv. (%)	Yield (%)	FE (%)	Fitness
1	Ni(OH)2/foam	KOH 1.10 M	11.1	1.410	25.0	99.0	95.2	96.2	0.890
2	Ni(OH)2/foam	KOH 0.80 M	12.1	1.410	25.4	99.0	95.2	96.2	0.874
3	Ni(OH)2/foam	KOH 0.83 M	8.2	1.411	25.2	99.0	95.4	95.0	0.866
4	Ni(OH)2/foam	KOH 0.84 M	11.9	1.412	26.1	99.2	95.4	95.1	0.855
5	Ni(OH)2/foam	KOH 0.45 M	11.7	1.415	25.0	99.2	94.8	94.6	0.854
6	Ni(OH)2/foam	KOH 0.65 M	8.8	1.412	25.7	99.2	94.9	94.7	0.851
7	Ni(OH)2/foam	KOH 0.42 M	12.0	1.410	25.2	99.2	94.3	94.6	0.851
8	Ni(OH)2/foam	KOH 0.69 M	11.0	1.416	25.0	99.2	94.9	94.7	0.850
9	Ni(OH)2/foam	KOH 0.71 M	11.5	1.412	25.5	99.2	94.5	94.9	0.844
10	Ni(OH)2/foam	KOH 0.43 M	11.8	1.412	25.4	99.2	94.8	93.2	0.837

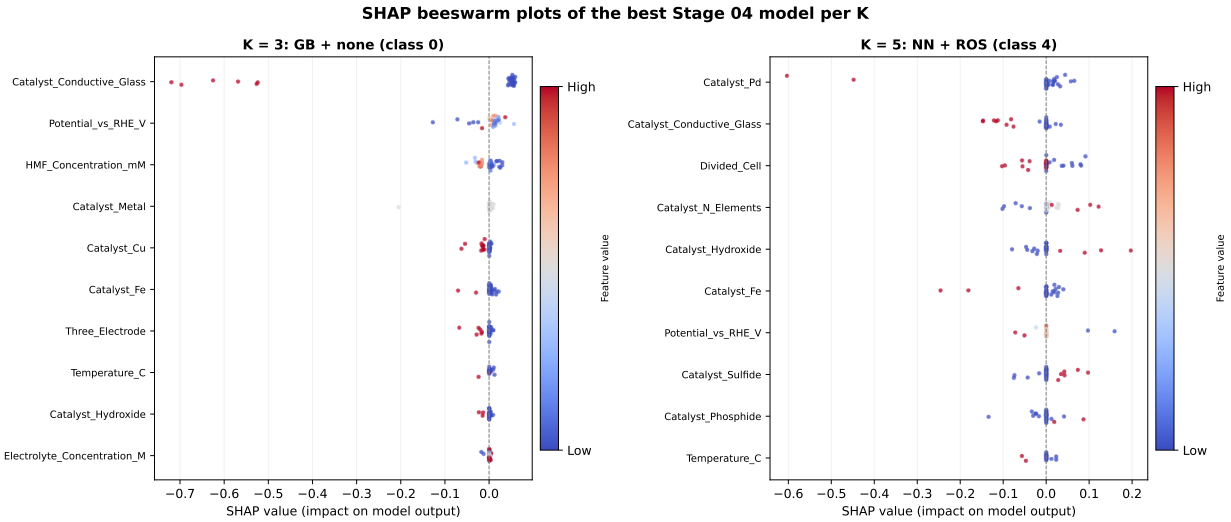


Figure 11: SHAP beeswarm plots for the majority class at $K = 3$ and $K = 5$, coloured by feature value.

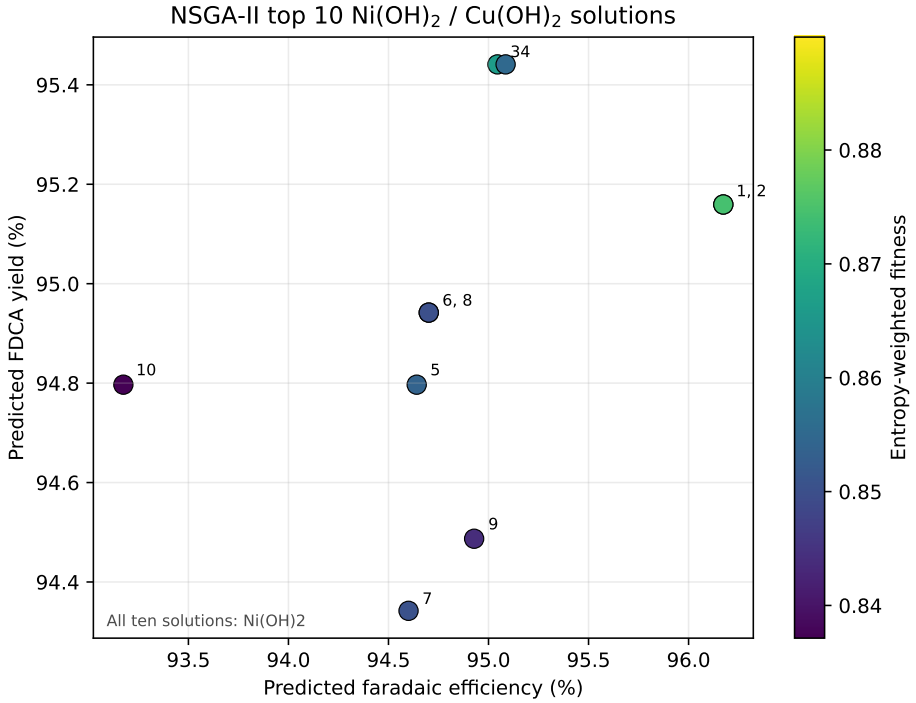


Figure 12: Predicted FDCA yield versus FE for the top-10 hydroxide candidates, coloured by entropy-weighted fitness.

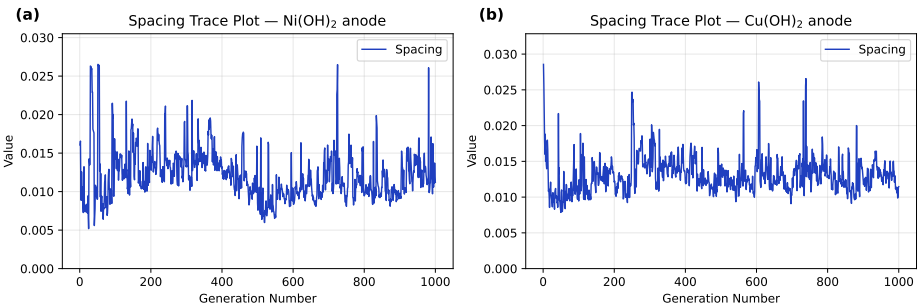


Figure 13: Spacing metric of the first Pareto front over generations.

Why 0.76 was not reached. Several individual repeats of the best binary model exceed 0.76, but the 25-repeat mean does not. Three factors cap performance: (i) only 119 records from fewer than one hundred papers, many sharing catalysts; (ii) a positive class that is the majority, so macro-F1 depends heavily on the 30 low-performance records; and (iii) between-paper differences in protocol that the descriptors cannot represent. Two routes would raise the reported number without improving the model: random (non-grouped) cross-validation, which lets near-duplicate records from one paper appear in training and test, and redefining the threshold or clusters to suit the classifier. We deliberately did neither.

Evaluation lessons. Silent exclusion of infeasible folds can make a model look several tenths better than it is; reporting feasibility explicitly is essential when oversampling tiny classes. Hyperparameters were selected on the same grouped protocol as the reported scores, which introduces a mild optimistic bias; a nested or external validation set would remove it.

Limitations. Outcomes were imputed for K-means only, so some cluster labels are pseudo-labels. Descriptors are parsed from free text and remain coarse (no loading, morphology or surface area). SHAP rankings and NSGA-II candidates are associational and model-dependent.

5 Conclusions

A leakage-aware pipeline for literature-scale electrocatalysis data reaches a publication-grouped macro-F1 of 0.731 for a $\geq$90% performance screen and 0.626 for three-cluster labels, well above chance but short of 0.76. Correct descriptor extraction and honest fold accounting mattered more than model choice. Expanding the corpus, reporting operating conditions completely and validating the NSGA-II candidates experimentally are the most promising next steps.

Data and code availability

All results, figures and this manuscript are regenerated by `python run_all.py`; this document is produced by `scripts/10_latex_report.py` from the files in `results/` and `figures/`.

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